Shi Luode

+86 188-8927-7075 | e1539282@u.nus.edu | starryneol.github.io/Personal_Web

Research Interests: Embodied AI and Robot Learning, Reinforcement Learning for Robotics, and Neural Combinatorial Optimization

EDUCATION

National University of Singapore (NUS) ***08/2025–Present***

Master of Science in Robotics | GPA: 4.07/5.00 | Expected graduation: Nov 28, 2026

- Research focus: autonomous exploration, reinforcement learning, decision optimization, viewpoint planning, and graph-based policy learning.
- Relevant coursework: Topics in Robotics (A+), AI Planning and Decision Making, Cognitive Robotics, Robot Dynamics and Control, and Machine Learning in Robotics.

Shandong University (SDU) ***09/2021–06/2025***

B.Eng. in Optical Information Science and Engineering | GPA: 3.61/5.00 | Rank: 7/43

RESEARCH EXPERIENCE

Preference-Adaptive Multi-Agent Optimization for Multi-Objective Combinatorial Optimization ***04/2026–Present***

Manuscript under review, MARMoT Lab, National University of Singapore

Authors: **Luode Shi***, Conghui Wang, Mingfeng Fan, and Guillaume Adrien Sartoretti*

- Developed PAMAO for multi-agent routing and scheduling, where agents must follow a shared objective trade-off while making role-specific decisions.
- Added agent-wise preference adaptation (AWPI) to MACSIM, PARCO-WECA, and DCAN variants. Collaborative Self-Improvement (CSI) compares solutions generated under the target and nearby preferences, then uses a strong solution for the target trade-off as a training example.
- Evaluated on generated and public benchmarks: PAMAO-M/P improved routing results over their respective backbones; PAMAO-M improved generated scheduling and public benchmark solution-set quality; PAMAO-D outperformed DCAN in most tested two- and three-objective settings under its official greedy protocol.

Ariadne: Reinforcement Learning for Single-Robot Autonomous Exploration ***12/2025–05/2026***

Research Project, MARMoT Laboratory, National University of Singapore

Advisor: Asst. Prof. Guillaume Sartoretti | Robotics, Autonomous Exploration, Reinforcement Learning

- Built a single-robot exploration system that selects observation viewpoints to reveal unknown areas in grid maps; the robot moves one A* step, updates its map, and replans at each decision.
- Clustered frontiers and sampled Gaussian viewpoints to create candidate actions; snapped candidates to traversable graph nodes, removed duplicates, ranked them by exploration utility, and capped the set at 25 per decision.
- Implemented a Transformer-pointer policy with twin-critic Soft Actor-Critic. In the expert-supervised setup, a ground-truth Gaussian-viewpoint planner selected frontier-covering viewpoints and ordered a route; an imitation reward aligned policy choices with expert viewpoints and frontier clusters.
- Integrated Ray-parallel rollouts and evaluated the best checkpoint on 100 test maps (up to 600 decisions per run), with a mean travel distance of 517.1 m. HEADER reports 529 ± 67 m on its 100-scenario test set (arXiv:2510.15679).

ADDITIONAL EXPERIENCE

Machine Vision Intern, Shandong Institutes of Industrial Technology ***07/2024–08/2024***

Industrial inspection and machine vision

- Built a production-line inspection workflow for open glass bottles, from camera calibration and image threshold setup through defect filtering and result reporting.
- Used MXVision and image-region size/shape rules to identify empty positions, inverted bottles, and bottles that had not been opened; tuned the checks with production samples.
- Processed inspection results with C# scripts and delivered bottle counts to the client; also assisted with defect-inspection tasks for Hikrobot products.

TECHNICAL SKILLS

- **Reinforcement Learning:** decision-process modeling, Soft Actor-Critic, twin value networks, reward design, expert-route imitation, and policies guided by objective trade-offs.
- **Multi-Agent Optimization:** multi-agent routing and job-shop scheduling; preference-guided neural solvers, agent-role adaptation, collaborative training, and solution-quality evaluation.
- **Robotics and Exploration:** occupancy-map exploration, frontier and viewpoint selection, graph-based planning, candidate-action generation, route coverage, and A*-based path planning.
- **Machine Learning Systems:** Python, PyTorch, Transformer models, Ray, Hydra, TensorBoard, Lightning/RL4CO, parallel simulation and training, and unit testing.
- **Development Tools:** AI-assisted coding tools for code reading, experiment scripting, and debugging.
- **Languages:** IELTS 6.5; CET-4 and CET-6.

HONORS AND AWARDS

Optoelectronics Alumni Scholarship, School of Information Science and Engineering, Shandong University ***2025***
Second-Class Academic Scholarship, Shandong University ***2024***
Outstanding Participant, SIOM Summer Camp, University of Chinese Academy of Sciences ***2024***
Second-Class Academic Scholarship, Shandong University ***2023***
Third-Class Academic Scholarship, Shandong University ***2022***